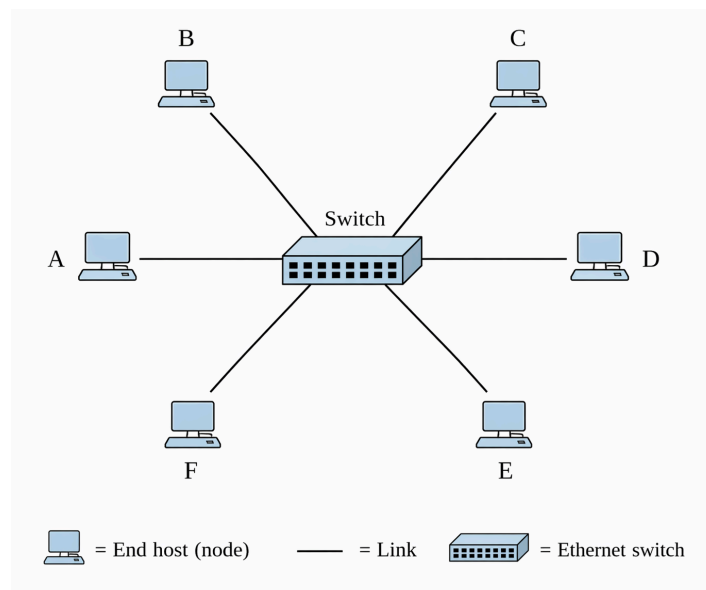


HW5_Subnets

1. (R11) Why is an ARP query sent within a broadcast frame? Why is an ARP response sent within a frame with a specific destination MAC address?

An ARP query is broadcast because the sender does not know the destination MAC address. An ARP response is unicast because the responder learns the sender's MAC from the query and can reply directly to save bandwidth.

2. (P26) Let's consider the operation of a learning switch in the context of a network in which nodes labeled A through F are star connected into an Ethernet switch. Suppose that (i) B sends a frame to E, (ii) E replies with a frame to B, (iii) A sends a frame to B, (iv) B replies with a frame to A. The switch table is initially empty. Show the state of the switch table before and after each of these events. For each of these events, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answers.



A. Before: Empty.

After: {B: Link B}.

Forwarded on Links A, C, D, E, and F because E's location is unknown and the frame must be flooded.

B. Before: {B: Link B}.

After: {B: Link B, E: Link E}.

Forwarded on Link B because B is already in the switch table.

C. Before: {B: Link B, E: Link E}.

After: {B: Link B, E: Link E, A: Link A}.

Forwarded on Link B because B is already in the switch table.

D. Before: {B: Link B, E: Link E, A: Link A}.

After: {B: Link B, E: Link E, A: Link A}.

Forwarded on Link A because A is already in the switch table.

3. (P31) In this problem, you will put together much of what you have learned about Internet protocols. Suppose you walk into a room, connect to Ethernet, and want to download a Web page. What are all the protocol steps that take place, starting from powering on your PC to getting the Web page? Assume there is nothing in our DNS or browser caches when you power on your PC. (*Hint:* The steps include the use of Ethernet, DHCP, ARP, DNS, TCP, and HTTP protocols.) Explicitly indicate in your steps how you obtain the IP and MAC addresses of a gateway router.

DHCP: The PC broadcasts a request to obtain an IP address, subnet mask, gateway IP, and DNS IP.

ARP: The PC broadcasts a query for the gateway's IP to find its MAC address.

DNS: The PC queries the DNS server to resolve the website's URL into an IP address.

TCP: The PC completes a 3-way handshake with the web server's IP to establish a connection. HTTP: The PC sends a GET request over the TCP connection to retrieve the webpage.

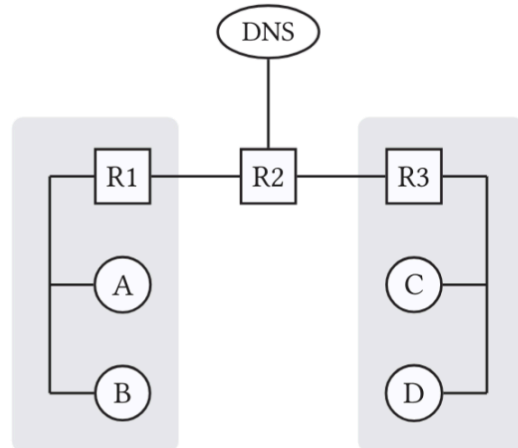
In this question, consider the network topology to the right.

A, B, and R1 are all in the same subnet, and are connected on a single shared medium (also called a bus).

Likewise, C, D, and R3 are all in the same subnet (different from the above subnet), and are connected on a single shared medium.

There is one DNS resolver in the network, labeled DNS. Assume everyone uses the same DNS resolver.

In this question, each subpart continues on from the previous one.



Q5.1 (2 points) Host A joins the network for the first time, with all caches empty and no active connections.

Host A broadcasts a DHCP Discover request. R1 sends a DHCP Offer reply containing some IP addresses. Host A accepts this offer.

Which IP addresses were contained in R1's DHCP Offer? Select all that apply.

- ☐ R1's IP ☐ R2's IP ☐ A's IP ☐ B's IP ☐ DNS's IP

Q5.2 (2 points) Host A now wants to send an HTTP request to Host C.

Before any further steps, select all fields that Host A currently knows how to fill in.

- ☐ Source IP ☐ Destination MAC ☐ None of the above
☐ Destination IP ☐ Source port
☐ Source MAC ☐ Destination port

Q5.3 (2 points) From the previous subpart, there are some fields that Host A currently does not know how to fill in.

Select all additional protocols that Host A needs to run in order to fill in those missing fields.

- ☐ ARP ☐ DHCP ☐ DNS ☐ TLS ☐ None

Eventually, A is able to fill in all fields, and sends a packet to C. The packet arrives at C.

Q5.4 (2 points) Select all devices that can deduce (possibly by reading packets they are not supposed to be reading) that Host A's IP address and Host C's IP address are communicating.

- ☐ B ☐ R1 ☐ R2 ☐ R3 ☐ D ☐ None

Q5.1: R1's IP, A's IP, DNS's IP.

Q5.2: Source IP, Source MAC, Source port, Destination port.

Q5.3: ARP, DNS. Q5.4: B, R1, R2, R3, D.

Q5.5 (2 points) At this point, which of these IP-to-MAC mappings exists in Host A's cache? Assume no entries have expired, and nobody has sent an unsolicited ARP message. Select all that apply.

- | | | |
|---|--|--|
| ☐ C's IP to C's MAC | ☐ DNS's IP to DNS's MAC | ☐ R2's IP to R1's MAC |
| ☐ C's IP to R1's MAC | ☐ DNS's IP to R1's MAC | ☐ B's IP to B's MAC |

At this point, Host A leaves and re-joins the network, which means that all of Host A's caches have been cleared.

Q5.6 (1 point) After Host A re-joins the network, does Host A still have the same IP address?

- ☐ Always ☐ Sometimes ☐ Never

For the rest of the question, assume that R1 is using NAT with Port Address Translation, as shown in lecture. Everyone in R1's subnet is using R1's public IP address, and A and B are assigned private IP addresses.

Q5.7 (1 point) Host A wants to communicate with Host C again.

At some point, Host A performs a DNS lookup for Host C's IP address.

Does R1 need to perform NAT translation on the DNS request packet?

- ☐ Yes, because Host A's IP address is private.
☐ Yes, because in NAT, the border router always modifies the source port for outgoing packets.
☐ No, because DNS messages fit in a single UDP packet, so no long-term connection is formed.
☐ No, because the DNS server's IP address is public.

Q5.8 (2 points) If Host A wants to send a packet to Host C, what additional steps does **Host A** need to do, now that NAT is enabled? Select all that apply.

- ☐ Choose a source port number that nobody else in the subnet is using.
☐ Choose a destination port number that nobody else in the subnet is using.
☐ Write R1's IP address in the source IP field.
☐ Write R1's IP address in the destination IP field.
☐ None of the above

Q5.9 (2 points) Now that NAT is enabled, select all devices that can deduce (possibly by reading packets they are not supposed to be reading) that Host A's IP address and Host C's IP address are communicating.

- ☐ B ☐ R1 ☐ R2 ☐ R3 ☐ DNS ☐ None

Q5.5: None.

Q5.6: Sometimes.

Q5.7: Yes, because Host A's IP address is private.

Q5.8: None of the above.

Q5.9: B, R1.